Parametric Equations & Modeling

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating parametric equations**

Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.

$x(2) = 4 - 1 = 3$. $y(2) = 4 + 3 = 7$. Point: $(3, 7)$.

2 **[Foundational]** • **Target: Evaluating parametric equations**

Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.

$x\left(\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{3}\right) = 3\left(\frac{1}{2}\right) = \frac{3}{2}$. $y\left(\frac{\pi}{3}\right) = 3\sin\left(\frac{\pi}{3}\right) = 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$. Since $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$, the curve traces a circle of radius 3 centered at the origin.

3 **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.

From $x = t + 1$: $t = x - 1$. Substitute: $y = (x - 1)^2 - 2 = x^2 - 2x + 1 - 2 = x^2 - 2x - 1$. This is a parabola.

4 **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.

From $x = 2t - 3$: $t = \frac{x+3}{2}$. Substitute: $y = 4\left(\frac{x+3}{2}\right) + 1 = 2(x + 3) + 1 = 2x + 7$. This is a line: $y = 2x + 7$.

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5 [Foundational] • Target: Visualizing a parametric curve's components

The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.

a Eliminate the parameter t to find the rectangular equation relating x and y for this curve. From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}$, so $y^2 = 4x$.

b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown). $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just x and y plotted individually against the parameter t.

(a) From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}$, so $y^2 = 4x$.; (b) $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just x and y plotted individually against the parameter t.

6 [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.

$x(t) = 1 + 4t$, $y(t) = 2 + 8t$ for $0 \le t \le 1$ (since the change in x is $5 - 1 = 4$ and the change in y is $10 - 2 = 8$). Check: at $t = 0$, $(1, 2)$; at $t = 1$, $(5, 10)$ ✓.

7 [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.

$x(t) = 5\cos t$, $y(t) = 3\sin t$, for $0 \le t < 2\pi$. Check: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1$, matching the standard ellipse equation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modeling projectile motion parametrically

A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)\,t$ and $y(t) = 40\sin(30^\circ)\,t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.

$x(2) = 40\cos(30^\circ)(2) \approx 40(0.866)(2) \approx 69.3$ m.
$y(2) = 40\sin(30^\circ)(2) - 4.9(4) = 40(0.5)(2) - 19.6 = 40 - 19.6 = 20.4$ m. Position: approximately $(69.3, 20.4)$.

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9 [Intermediate] • Target: Modeling projectile motion parametrically

For the same projectile, find the time at which the ball lands (returns to $y = 0$).

Set $y(t) = 0$: $20t - 4.9t^2 = 0$, so $t(20 - 4.9t) = 0$, giving $t = 0$ or $t = \frac{20}{4.9} \approx 4.08$ seconds. The ball lands at $t \approx 4.08$ s.

10 [Intermediate] • Target: Comparing parametric speed and direction

Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.

Both satisfy $y = x^2$ (for B: $y = 4t^2 = (2t)^2 = x^2$ ✓), so they trace identical curves. But object B's x-coordinate changes twice as fast per unit of t as object A's, meaning B moves through the same path twice as fast — the parametrization encodes speed and direction of travel, which the rectangular equation $y = x^2$ alone does not capture.

11 [Intermediate] • Target: Restricting the parameter's domain

The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in [0, \frac{\pi}{2}]$.

Restricting to $t \in [0, \frac{\pi}{2}]$ traces only the arc from $(1, 0)$ (at $t = 0$) to $(0, 1)$ (at $t = \frac{\pi}{2}$) — the quarter of the circle lying in Quadrant I.

12 [Intermediate] • Target: Restricting the parameter's domain

Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

Eliminating the parameter gives $x = y^2$ for all t, but since $y = t$ directly, restricting $t \geq 0$ forces $y \geq 0$ as well — so only the upper half of the sideways parabola (where $y \geq 0$) is actually traced, even though the unrestricted equation $x = y^2$ would include the lower half too.

13 [Intermediate] • Target: Parametric equations for circular motion with a phase shift

A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.

At $t = 0$: $x(0) = 4\cos\left(\frac{\pi}{4}\right) = 4\left(\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$, $y(0) = 4\sin\left(\frac{\pi}{4}\right) = 2\sqrt{2}$. Without the phase shift, the point would start at $(4, 0)$ when $t = 0$; the added $\frac{\pi}{4}$ advances the starting position by an eighth of a full revolution counterclockwise, to $(2\sqrt{2}, 2\sqrt{2})$, while the circle traced and the speed of travel remain unchanged.

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14 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x.

Final Answer: $y = -3x - 21$. From $x = t - 6$: $t = x + 6$. Substitute: $y = -3(x + 6) - 3 = -3x - 21$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Eliminate the parameter from $x(t) = t + 9$ and $y(t) = 3t + 4$ to find the rectangular equation for y in terms of x.

Final Answer: $y = 3x - 23$. From $x = t + 9$: $t = x - 9$. Substitute: $y = 3(x - 9) + 4 = 3x - 23$.

16 [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Eliminate the parameter from $x(t) = t$ and $y(t) = 8t - 9$ to find the rectangular equation for y in terms of x.

Final Answer: $y = 8x - 9$. From $x = t$: $t = x$. Substitute: $y = 8(x) - 9 = 8x - 9$.

17 [Challenging] • Target: Modeling projectile motion parametrically • Cross-Lesson: 4.01 Inverse & Composite Functions

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -8t + 3$ to find the rectangular equation for y in terms of x.

Final Answer: $y = -8x - 13$. From $x = t - 2$: $t = x + 2$. Substitute: $y = -8(x + 2) + 3 = -8x - 13$.

18 [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -t - 5$ to find the rectangular equation for y in terms of x.

Final Answer: $y = -x - 7$. From $x = t - 2$: $t = x + 2$. Substitute: $y = -1(x + 2) - 5 = -x - 7$.

19 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Given $x(t) = t^2 + 6$ and $y(t) = -2t + 8$, find the point on the curve at $t = 0$.

Final Answer: $(6, 8)$. $x(0) = (0)^2 + 6 = 6$. $y(0) = -2(0) + 8 = 8$. Point: $(6, 8)$.

20 [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Given $x(t) = t^2 + 2$ and $y(t) = -2t + 7$, find the point on the curve at $t = -1$.

Final Answer: $(3, 9)$. $x(-1) = (-1)^2 + 2 = 3$. $y(-1) = -2(-1) + 7 = 9$. Point: $(3, 9)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = -3t - 2$ and $y(t) = -5t$, find the point on the curve at $t = -2$.

a. $(10, 4)$

b. $(4, 10)$ ✓

c. $(4, 11)$

d. $(5, 10)$

$x(-2) = -3(-2) - 2 = 4$. $y(-2) = -5(-2) = 10$. Point: $(4, 10)$.

22 [Foundational] • Target: Eliminating the parameter

Given $x(t) = -5t - 5$ and $y(t) = 2t - 3$, find the point on the curve at $t = 2$.

a. $(-15, 2)$

b. $(-14, 1)$

c. $(-15, 1)$ ✓

d. $(1, -15)$

$x(2) = -5(2) - 5 = -15$. $y(2) = 2(2) - 3 = 1$. Point: $(-15, 1)$.

23 [Foundational] • Target: Visualizing a parametric curve's components

Given $x(t) = -2t + 9$ and $y(t) = 3t + 5$, find the point on the curve at $t = 0$.

a. $(9, 5)$ ✓

b. $(9, 6)$

c. $(10, 5)$

d. $(5, 9)$

$x(0) = -2(0) + 9 = 9$. $y(0) = 3(0) + 5 = 5$. Point: $(9, 5)$.

24 [Foundational] • Target: Parametric equations for lines and conics

Given $x(t) = -3t + 4$ and $y(t) = -3t + 4$, find the point on the curve at $t = -3$.

a. $(13, 13)$ ✓

b. $(11, 15)$

c. $(14, 13)$

d. $(13, 14)$

$x(-3) = -3(-3) + 4 = 13$. $y(-3) = -3(-3) + 4 = 13$. Point: $(13, 13)$.

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25 [Foundational] • Target: Modeling projectile motion parametrically

Given $x(t) = 4t - 7$ and $y(t) = t - 7$, find the point on the curve at $t = -5$.

a. $(-27, -12)$ ✓

b. $(-26, -12)$

c. $(-27, -11)$

d. $(-12, -27)$

$x(-5) = 4(-5) - 7 = -27$. $y(-5) = 1(-5) - 7 = -12$. Point: $(-27, -12)$.

26 [Foundational] • Target: Comparing parametric speed and direction

Given $x(t) = 6t + 9$ and $y(t) = 2t + 5$, find the point on the curve at $t = -2$.

a. $(-2, 1)$

b. $(-3, 1)$ ✓

c. $(-3, 2)$

d. $(1, -3)$

$x(-2) = 6(-2) + 9 = -3$. $y(-2) = 2(-2) + 5 = 1$. Point: $(-3, 1)$.

27 [Foundational] • Target: Restricting the parameter's domain

Given $x(t) = 4t + 4$ and $y(t) = t + 9$, find the point on the curve at $t = -6$.

a. $(-20, 4)$

b. $(-19, 3)$

c. $(3, -20)$

d. $(-20, 3)$ ✓

$x(-6) = 4(-6) + 4 = -20$. $y(-6) = 1(-6) + 9 = 3$. Point: $(-20, 3)$.

28 [Foundational] • Target: Parametric equations for circular motion with a phase shift

Given $x(t) = 6t - 4$ and $y(t) = 5t - 3$, find the point on the curve at $t = 5$.

a. $(27, 22)$

b. $(22, 26)$

c. $(26, 22)$ ✓

d. $(26, 23)$

$x(5) = 6(5) - 4 = 26$. $y(5) = 5(5) - 3 = 22$. Point: $(26, 22)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Evaluating Parametric Equations and Graph Points

Eliminate the parameter from $x(t) = t - 9$ and $y(t) = -7t + 7$ to find the rectangular equation for y in terms of x.

a. $y = 7x - 56$

b. $y = -7x - 55$

c. $y = -7x + 7$

d. $y = -7x - 56$ ✓

From $x = t - 9$: $t = x + 9$. Substitute: $y = -7(x + 9) + 7 = -7x - 56$.

30 [Intermediate] • Target: Eliminating the Parameter

Eliminate the parameter from $x(t) = t + 5$ and $y(t) = 7t - 2$ to find the rectangular equation for y in terms of x.

a. $y = 7x - 36$

b. $y = 7x - 37$ ✓

c. $y = 7x - 2$

d. $y = -7x - 37$

From $x = t + 5$: $t = x - 5$. Substitute: $y = 7(x - 5) - 2 = 7x - 37$.

31 [Intermediate] • Target: Projectile Motion Modeling

Eliminate the parameter from $x(t) = t - 1$ and $y(t) = 8t - 5$ to find the rectangular equation for y in terms of x.

a. $y = 8x - 5$

b. $y = -8x + 3$

c. $y = 8x + 3$ ✓

d. $y = 8x + 4$

From $x = t - 1$: $t = x + 1$. Substitute: $y = 8(x + 1) - 5 = 8x + 3$.

32 [Intermediate] • Target: Direction of Motion and Speed

Eliminate the parameter from $x(t) = t - 5$ and $y(t) = t - 7$ to find the rectangular equation for y in terms of x.

a. $y = -x - 2$

b. $y = x - 7$

c. $y = x - 2$ ✓

d. $y = x - 1$

From $x = t - 5$: $t = x + 5$. Substitute: $y = 1(x + 5) - 7 = x - 2$.

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33 [Intermediate] • Target: Evaluating parametric equations

Eliminate the parameter from $x(t) = t + 6$ and $y(t) = t + 4$ to find the rectangular equation for y in terms of x.

a. $y = x - 2$ ✓

b. $y = -x - 2$

c. $y = x - 1$

d. $y = x + 4$

From $x = t + 6$: $t = x - 6$. Substitute: $y = 1(x - 6) + 4 = x - 2$.

34 [Intermediate] • Target: Eliminating the parameter

Given $x(t) = t^2 - 9$ and $y(t) = 4t + 3$, find the point on the curve at $t = -1$.

a. $(-8, 1)$

b. $(-1, -8)$

c. $(-6, -1)$

d. $(-8, -1)$ ✓

$x(-1) = (-1)^2 - 9 = -8$. $y(-1) = 4(-1) + 3 = -1$. Point: $(-8, -1)$.

35 [Intermediate] • Target: Visualizing a parametric curve's components

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ (330°).

a. $(2\sqrt{3}, 2)$

b. $(2\sqrt{3}, -2)$ ✓

c. $(-2, 2\sqrt{3})$

d. $(-2\sqrt{3}, -2)$

$x = 4\cos\left(\frac{11\pi}{6}\right) = 2\sqrt{3}$. $y = 4\sin\left(\frac{11\pi}{6}\right) = -2$. Point: $(2\sqrt{3}, -2)$.

36 [Intermediate] • Target: Parametric equations for lines and conics

A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{3\pi}{2}$ (270°).

a. $(0, -8)$ ✓

b. $(-8, 0)$

c. $(-4\sqrt{3}, -4)$

d. $(0, 8)$

$x = 8\cos\left(\frac{3\pi}{2}\right) = 0$. $y = 8\sin\left(\frac{3\pi}{2}\right) = -8$. Point: $(0, -8)$.

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37 [Intermediate] • Target: Modeling projectile motion parametrically

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{7\pi}{4}$ (315°).

a. $(2\sqrt{2}, 2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(-2\sqrt{2}, -2\sqrt{2})$

d. $(2\sqrt{2}, -2\sqrt{2})$ ✓

$x = 4\cos\left(\frac{7\pi}{4}\right) = 2\sqrt{2}$. $y = 4\sin\left(\frac{7\pi}{4}\right) = -2\sqrt{2}$. Point: $(2\sqrt{2}, -2\sqrt{2})$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A projectile is launched with initial speed 49 m/s at 60° above horizontal, so $y(t) = 49\sin(60^\circ)\,t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 65.3 ✓

b. 70.3

c. 67.3

d. 63.3

$y(2) = 49\sin(60^\circ)(2) - 4.9(2)^2 \approx 49(0.86603)(2) - 4.9(4) \approx 65.3$ m.

39 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A projectile is launched with initial speed 60 m/s at 30° above horizontal, so $y(t) = 60\sin(30^\circ)\,t - 4.9t^2$. Find $y(1)$, to one decimal place.

a. 27.1

b. 25.1 ✓

c. 23.1

d. 30.1

$y(1) = 60\sin(30^\circ)(1) - 4.9(1)^2 \approx 60(0.5)(1) - 4.9(1) \approx 25.1$ m.

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40 [Challenging] • Target: Evaluating Parametric Equations and Graph Points • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A particle moves according to $x(t) = t^2 - 3$ and $y(t) = 2t + 1$. What is its position at $t = 3$?

a. $(6, 7)$ ✓

b. $(9, 7)$

c. $(6, 6)$

d. $(3, 7)$

$x(3) = 3^2 - 3 = 6$. $y(3) = 2(3) + 1 = 7$. The position point is $(6, 7)$.

41 [Challenging] • Target: Evaluating Parametric Equations and Graph Points • Cross-Lesson: 1.03 Rates of Change & Concavity

Given $x(t) = 4\cos t$ and $y(t) = 4\sin t$ for $0 \le t \le 2\pi$, what curve is traced and in which direction?

a. **A circle of radius 4 traced counterclockwise** ✓

b. A circle of radius 4 traced clockwise

c. An ellipse of semi-major axis 4

d. A line segment from $(-4, -4)$ to $(4, 4)$

At $t = 0$, position is $(4, 0)$. At $t = \frac{\pi}{2}$, position is $(0, 4)$. As t increases, motion proceeds counterclockwise around a circle of radius 4.

42 [Challenging] • Target: Eliminating the Parameter • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Eliminate the parameter from $x = 3\cos\theta$ and $y = 5\sin\theta$.

a. $\frac{x^2}{9} + \frac{y^2}{25} = 1$ ✓

b. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

c. $x^2 + y^2 = 34$

d. $9x^2 + 25y^2 = 1$

Isolate trig functions: $\cos\theta = \frac{x}{3}$ and $\sin\theta = \frac{y}{5}$. Using $\cos^2\theta + \sin^2\theta = 1$ yields $\frac{x^2}{9} + \frac{y^2}{25} = 1$.

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43 [Challenging] • Target: Projectile Motion Modeling • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A projectile is launched with initial speed $v_0 = 80$ ft/s at an angle of 30° from the horizontal from ground level. Which parametric equations model its path ($g = 32$ ft/s^2)?

a. $x(t) = 40\sqrt{3}t$ **and** $y(t) = 40t - 16t^2$ ✓

b. $x(t) = 40t$ and $y(t) = 40\sqrt{3}t - 16t^2$

c. $x(t) = 80t$ and $y(t) = 80t - 32t^2$

d. $x(t) = 40\sqrt{3}t$ and $y(t) = 40t - 32t^2$

$v_{0x} = 80\cos(30^\circ) = 80\left(\frac{\sqrt{3}}{2}\right) = 40\sqrt{3}$. $v_{0y} = 80\sin(30^\circ) = 80\left(\frac{1}{2}\right) = 40$. In feet, $y(t) = v_{0y}t - \frac{1}{2}gt^2 = 40t - 16t^2$.

44 [Challenging] • Target: Projectile Motion Modeling • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

What is the maximum height attained by a projectile modeled by $y(t) = 64t - 16t^2$?

a. 64 ft ✓

b. 32 ft

c. 128 ft

d. 48 ft

Maximum height occurs at vertex $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ s. Maximum height: $y(2) = 64(2) - 16(4) = 128 - 64 = 64$ ft.

45 [Challenging] • Target: Direction of Motion and Speed • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

If a curve is given by $x(t) = t + 1$ and $y(t) = t^2$, what is the average rate of change $\frac{\Delta y}{\Delta x}$ from $t = 1$ to $t = 3$?

a. 4 ✓

b. 8

c. 2

d. 6

At $t = 1$: $(2, 1)$. At $t = 3$: $(4, 9)$. $\frac{\Delta y}{\Delta x} = \frac{9-1}{4-2} = \frac{8}{2} = 4$.

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